# Hex Type D Flip-Flop

# MC14174B

The MC14174B hex type D flip-flop is constructed with MOS P-channel and N-channel enhancement mode devices in a single monolithic structure. Data on the D inputs which meets the setup time requirements is transferred to the Q outputs on the positive edge of the clock pulse. All six flip-flops share common clock and reset inputs. The reset is active low, and independent of the clock.

### Features

- Static Operation
- All Inputs and Outputs Buffered
- Diode Protection on All Inputs
- Supply Voltage Range = 3.0 Vdc to 18 Vdc
- Capable of Driving Two Low-Power TTL Loads or One Low-Power Schottky TTL Load Over the Rated Temperature Range
- Functional Equivalent to TTL 74174
- These Devices are Pb-Free and are RoHS Compliant
- NLV Prefix for Automotive and Other Applications Requiring Unique Site and Control Change Requirements; AEC-Q100 Qualified and PPAP Capable

**MAXIMUM RATINGS** (Voltages Referenced to $V_{SS}$)

| Parameter | Symbol | Value | Unit |
|---|---|---|---|
| DC Supply Voltage Range | $V_{DD}$ | −0.5 to +18.0 | V |
| Input or Output Voltage Range (DC or Transient) | $V_{in}$, $V_{out}$ | −0.5 to $V_{DD}$ + 0.5 | V |
| Input or Output Current (DC or Transient) per Pin | $I_{in}$, $I_{out}$ | ±10 | mA |
| Power Dissipation, per Package (Note 2) | $P_D$ | 500 | mW |
| Ambient Temperature Range | $T_A$ | −55 to +125 | °C |
| Storage Temperature Range | | −65 to +150 | °C |
| Lead Temperature (8-Second Soldering) | | 260 | °C |

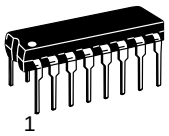
Stresses exceeding those listed in the Maximum Ratings table may damage the device. If any of these limits are exceeded, device functionality should not be assumed, damage may occur and reliability may be affected.

2. Temperature Derating: Plastic “P and D/DW”
Packages: − 7.0 mW/°C From 65 °C To 125 °C

This device contains protection circuitry to guard against damage due to high static voltages or electric fields. However, precautions must be taken to avoid applications of any voltage higher than maximum rated voltages to this high-impedance circuit. For proper operation, $V_{in}$ and $V_{out}$ should be constrained to the range $V_{SS} \le (V_{in}$ or $V_{out}) \le V_{DD}$.

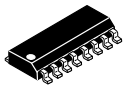
Unused inputs must always be tied to an appropriate logic voltage level (e.g., either $V_{SS}$ or $V_{DD}$). Unused outputs must be left open.

**MARKING DIAGRAMS**



**PDIP-16**
**P SUFFIX**
**CASE 648**



**SOIC-16**
**D SUFFIX**
**CASE 751B**




A = Assembly Location
WL = Wafer Lot
YY, Y = Year
WW = Work Week
G = Pb-Free Package

**ORDERING INFORMATION**

| Device | Package | Shipping† |
|---|---|---|
| MC14174BDR2G | SOIC-16 (Pb-Free) | 2500/Tape & Reel |
| NLV14174BDR2G | SOIC-16 (Pb-Free) | 2500/Tape & Reel |

**DISCONTINUED** (Note 1)

| | | |
|---|---|---|
| MC14174BCPG | PDIP-16 (Pb-Free) | 500 Units/Rail |

† For information on tape and reel specifications, including part orientation and tape sizes, please refer to our Tape and Reel Packaging Specification Brochure, BRD8011/D.

1. **DISCONTINUED:** This device is not recommended for new design. Please contact your **onsemi** representative for information. The most current information on this device may be available on www.onsemi.com.

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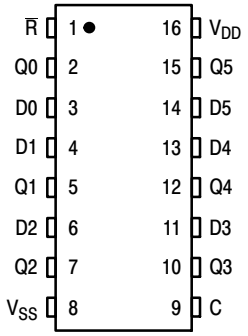
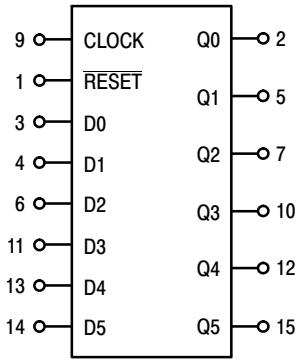



**Figure 1. Pin Assignment**




$V_{DD}$ = PIN 16
$V_{SS}$ = PIN 8

**Figure 2. Block Diagram**

**TRUTH TABLE (Positive Logic)**

| Inputs | | | Output | |
|---|---|---|---|---|
| Clock | Data | $\overline{\text{Reset}}$ | Q | |
| ↗ | 0 | 1 | 0 | |
| ↗ | 1 | 1 | 1 | |
| ↘ | X | 1 | Q | No Change |
| X | X | 0 | 0 | |

X = Don't Care

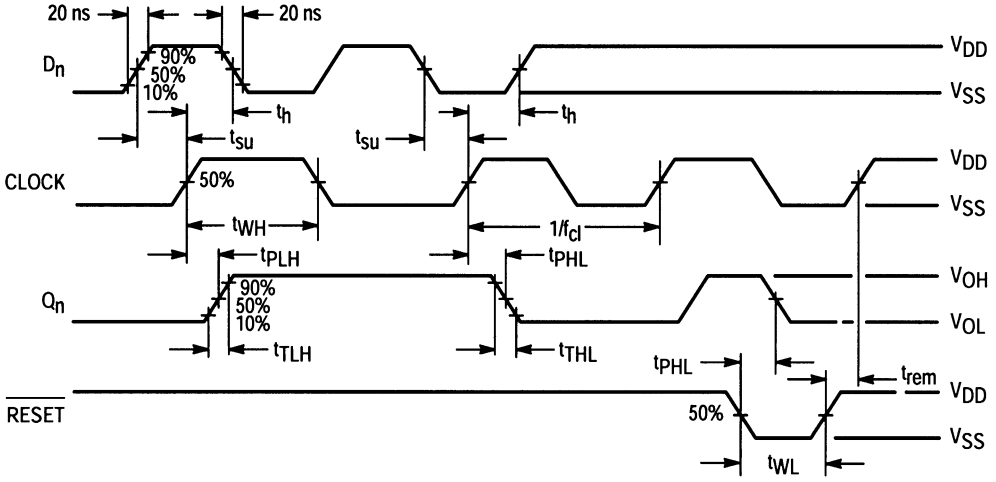



**Figure 3. Timing Diagram**

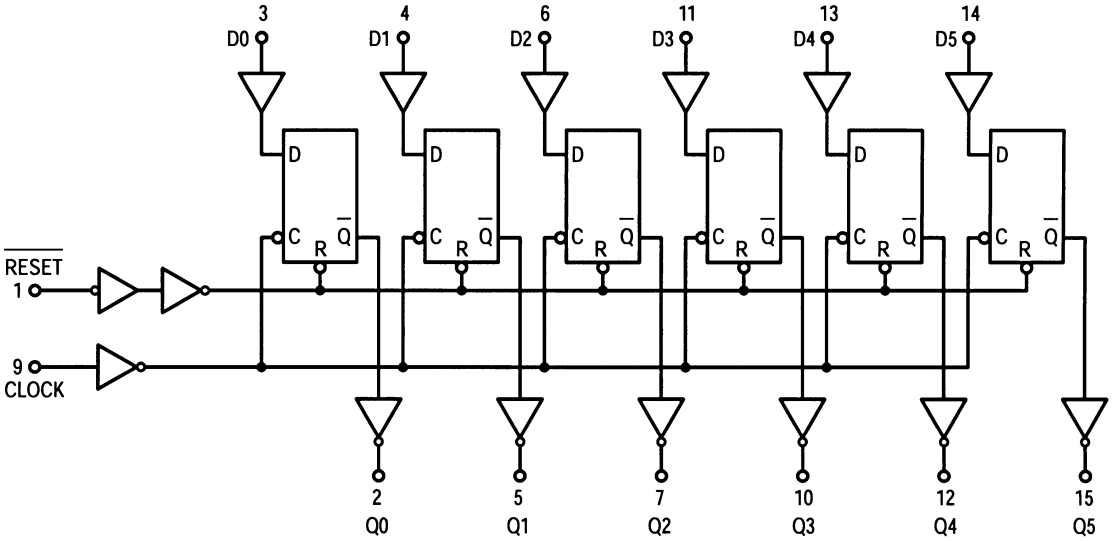



**Figure 4. Functional Block Diagram**



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**ELECTRICAL CHARACTERISTICS** (Voltages Referenced to $V_{SS}$)

| Characteristic | Sym-bol | $V_{DD}$ Vdc | −55 °C Min | −55 °C Max | 25 °C Min | 25 °C Typ (Note 3) | 25 °C Max | 125 °C Min | 125 °C Max | Unit |
|---|---|---|---|---|---|---|---|---|---|---|
| Output Voltage “0” Level<br>$V_{in}$ = $V_{DD}$ or 0 | $V_{OL}$ | 5.0<br>10<br>15 | –<br>–<br>– | 0.05<br>0.05<br>0.05 | –<br>–<br>– | 0<br>0<br>0 | 0.05<br>0.05<br>0.05 | –<br>–<br>– | 0.05<br>0.05<br>0.05 | Vdc |
| “1” Level<br>$V_{in}$ = 0 or $V_{DD}$ | $V_{OH}$ | 5.0<br>10<br>15 | 4.95<br>9.95<br>14.95 | –<br>–<br>– | 4.95<br>9.95<br>14.95 | 5.0<br>10<br>15 | –<br>–<br>– | 4.95<br>9.95<br>14.95 | –<br>–<br>– | Vdc |
| Input Voltage “0” Level<br>($V_O$ = 4.5 or 0.5 Vdc)<br>($V_O$ = 9.0 or 1.0 Vdc)<br>($V_O$ = 13.5 or 1.5 Vdc) | $V_{IL}$ | 5.0<br>10<br>15 | –<br>–<br>– | 1.5<br>3.0<br>4.0 | –<br>–<br>– | 2.25<br>4.50<br>6.75 | 1.5<br>3.0<br>4.0 | –<br>–<br>– | 1.5<br>3.0<br>4.0 | Vdc |
| “1” Level<br>($V_O$ = 0.5 or 4.5 Vdc)<br>($V_O$ = 1.0 or 9.0 Vdc)<br>($V_O$ = 1.5 or 13.5 Vdc) | $V_{IH}$ | 5.0<br>10<br>15 | 3.5<br>7.0<br>11 | –<br>–<br>– | 3.5<br>7.0<br>11 | 2.75<br>5.50<br>8.25 | –<br>–<br>– | 3.5<br>7.0<br>11 | –<br>–<br>– | Vdc |
| Output Drive Current<br>($V_{OH}$ = 2.5 Vdc) Source<br>($V_{OH}$ = 4.6 Vdc)<br>($V_{OH}$ = 9.5 Vdc)<br>($V_{OH}$ = 13.5 Vdc) | $I_{OH}$ | 5.0<br>5.0<br>10<br>15 | – 3.0<br>– 0.64<br>– 1.6<br>– 4.2 | –<br>–<br>–<br>– | – 2.4<br>– 0.51<br>– 1.3<br>– 3.4 | – 4.2<br>– 0.88<br>– 2.25<br>– 8.8 | –<br>–<br>–<br>– | – 1.7<br>– 0.36<br>– 0.9<br>– 2.4 | –<br>–<br>–<br>– | mAdc |
| ($V_{OL}$ = 0.4 Vdc) Sink<br>($V_{OL}$ = 0.5 Vdc)<br>($V_{OL}$ = 1.5 Vdc) | $I_{OL}$ | 5.0<br>10<br>15 | 0.64<br>1.6<br>4.2 | –<br>–<br>– | 0.51<br>1.3<br>3.4 | 0.88<br>2.25<br>8.8 | –<br>–<br>– | 0.36<br>0.9<br>2.4 | –<br>–<br>– | mAdc |
| Input Current | $I_{in}$ | 15 | – | ± 0.1 | – | ±0.00001 | ± 0.1 | – | ± 1.0 | μAdc |
| Input Capacitance, ($V_{in}$ = 0) | $C_{in}$ | – | – | – | – | 5.0 | 7.5 | – | – | pF |
| Quiescent Current<br>(Per Package) | $I_{DD}$ | 5.0<br>10<br>15 | –<br>–<br>– | 5.0<br>10<br>20 | –<br>–<br>– | 0.005<br>0.010<br>0.015 | 5.0<br>10<br>20 | –<br>–<br>– | 150<br>300<br>600 | μAdc |
| Total Supply Current (Note 4, 5)<br>(Dynamic plus Quiescent, Per Package)<br>($C_L$ = 50 pF on all outputs, all buffers switching) | $I_T$ | 5.0<br>10<br>15 | $I_T$ = (1.1 μA/kHz) f + $I_{DD}$<br>$I_T$ = (2.3 μA/kHz) f + $I_{DD}$<br>$I_T$ = (3.7 μA/kHz) f + $I_{DD}$ | | | | | | | μAdc |

Product parametric performance is indicated in the Electrical Characteristics for the listed test conditions, unless otherwise noted. Product performance may not be indicated by the Electrical Characteristics if operated under different conditions.

3. Data labelled “Typ” is not to be used for design purposes but is intended as an indication of the IC’s potential performance.
4. The formulas given are for the typical characteristics only at 25 °C.
5. To calculate total supply current at loads other than 50 pF: $I_T(C_L) = I_T(50\ pF) + (C_L - 50)$ Vfk where: $I_T$ is in μA (per package), $C_L$ in pF, V = ($V_{DD}$ − $V_{SS}$) in volts, f in kHz is input frequency, and k = 0.003.



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**SWITCHING CHARACTERISTICS** (Note 6) ($C_L$ = 50 pF, $T_A$ = 25 °C)

| Characteristic | Symbol | $V_{DD}$ Vdc | All Types | | | Unit |
|---|---|---|---|---|---|---|
| | | | Min | Typ (Note 7) | Max | |
| Output Rise and Fall Time<br>$t_{TLH}$, $t_{THL}$ = (1.35 ns/pF) $C_L$ + 32 ns<br>$t_{TLH}$, $t_{THL}$ = (0.6 ns/pF) $C_L$ + 20 ns<br>$t_{TLH}$, $t_{THL}$ = (0.4 ns/pF) $C_L$ + 20 ns | $t_{TLH}$, $t_{THL}$ | <br>5.0<br>10<br>15 | <br>–<br>–<br>– | <br>100<br>50<br>40 | <br>200<br>100<br>80 | ns |
| Propagation Delay Time — Clock to Q<br>$t_{PLH}$, $t_{PHL}$ = (0.9 ns/pF) $C_L$ + 165 ns<br>$t_{PLH}$, $t_{PHL}$ = (0.36 ns/pF) $C_L$ + 64 ns<br>$t_{PLH}$, $t_{PHL}$ = (0.26 ns/pF) $C_L$ + 52 ns | $t_{PLH}$, $t_{PHL}$ | <br>5.0<br>10<br>15 | <br>–<br>–<br>– | <br>210<br>85<br>65 | <br>400<br>160<br>120 | ns |
| Propagation Delay Time — Reset to Q<br>$t_{PHL}$ = (0.9 ns/pF) $C_L$ + 205 ns<br>$t_{PHL}$ = (0.36 ns/pF) $C_L$ + 79 ns<br>$t_{PHL}$ = (0.26 ns/pF) $C_L$ + 62 ns | $t_{PHL}$ | <br>5.0<br>10<br>15 | <br>–<br>–<br>– | <br>250<br>100<br>75 | <br>500<br>200<br>150 | ns |
| Clock Pulse Width | $t_{WH}$ | 5.0<br>10<br>15 | 150<br>90<br>70 | 75<br>45<br>35 | –<br>–<br>– | ns |
| $\overline{\text{Reset}}$ Pulse Width | $t_{WL}$ | 5.0<br>10<br>15 | 200<br>100<br>80 | 100<br>50<br>40 | –<br>–<br>– | ns |
| Clock Pulse Frequency | $f_{cl}$ | 5.0<br>10<br>15 | –<br>–<br>– | 7.0<br>12<br>15.5 | 2.0<br>5.0<br>6.5 | mHz |
| Clock Pulse Rise and Fall Time | $t_{TLH}$, $t_{THL}$ | 5.0<br>10<br>15 | –<br>–<br>– | –<br>–<br>– | 15<br>5.0<br>4.0 | μs |
| Data Setup Time | $t_{su}$ | 5.0<br>10<br>15 | 40<br>20<br>15 | 20<br>10<br>0 | –<br>–<br>– | ns |
| Data Hold Time | $t_h$ | 5.0<br>10<br>15 | 80<br>40<br>30 | 40<br>20<br>15 | –<br>–<br>– | ns |
| $\overline{\text{Reset}}$ Removal Time | $t_{rem}$ | 5.0<br>10<br>15 | 250<br>100<br>80 | 125<br>50<br>40 | –<br>–<br>– | ns |

6. The formulas given are for the typical characteristics only at 25°C.
7. Data labelled “Typ” is not to be used for design purposes but is intended as an indication of the IC’s potential performance.


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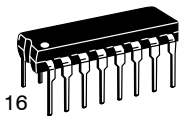
## REVISION HISTORY

| Revision | Description of Changes | Date |
|---|---|---|
| 9 | Rebranded the Data Sheet to **onsemi** format. MC14174BCPG OPN Marked as Discontinued. | 8/27/2025 |

This document has undergone updates prior to the inclusion of this revision history table. The changes tracked here only reflect updates made on the noted approval dates.



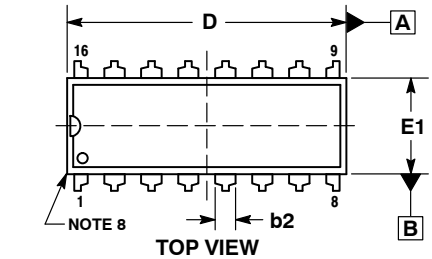
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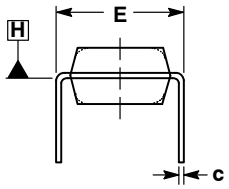

SCALE 1:1

**PDIP−16**
CASE 648−08
ISSUE V

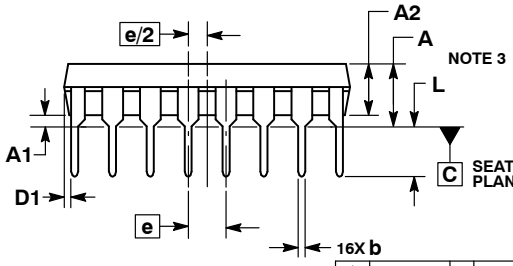
DATE 22 APR 2015



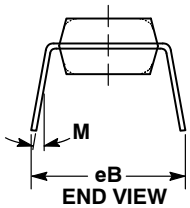

TOP VIEW




END VIEW
WITH LEADS CONSTRAINED
NOTE 5




SIDE VIEW




END VIEW
NOTE 6

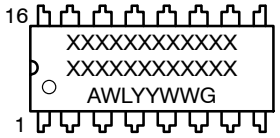
NOTES:
1. DIMENSIONING AND TOLERANCING PER ASME Y14.5M, 1994.
2. CONTROLLING DIMENSION: INCHES.
3. DIMENSIONS A, A1 AND L ARE MEASURED WITH THE PACKAGE SEATED IN JEDEC SEATING PLANE GAUGE GS−3.
4. DIMENSIONS D, D1 AND E1 DO NOT INCLUDE MOLD FLASH OR PROTRUSIONS. MOLD FLASH OR PROTRUSIONS ARE NOT TO EXCEED 0.10 INCH.
5. DIMENSION E IS MEASURED AT A POINT 0.015 BELOW DATUM PLANE H WITH THE LEADS CONSTRAINED PERPENDICULAR TO DATUM C.
6. DIMENSION eB IS MEASURED AT THE LEAD TIPS WITH THE LEADS UNCONSTRAINED.
7. DATUM PLANE H IS COINCIDENT WITH THE BOTTOM OF THE LEADS, WHERE THE LEADS EXIT THE BODY.
8. PACKAGE CONTOUR IS OPTIONAL (ROUNDED OR SQUARE CORNERS).

| | INCHES | | MILLIMETERS | |
|---|---|---|---|---|
| DIM | MIN | MAX | MIN | MAX |
| A | ---- | 0.210 | --- | 5.33 |
| A1 | 0.015 | ---- | 0.38 | --- |
| A2 | 0.115 | 0.195 | 2.92 | 4.95 |
| b | 0.014 | 0.022 | 0.35 | 0.56 |
| b2 | 0.060 TYP | | 1.52 TYP | |
| C | 0.008 | 0.014 | 0.20 | 0.36 |
| D | 0.735 | 0.775 | 18.67 | 19.69 |
| D1 | 0.005 | ---- | 0.13 | --- |
| E | 0.300 | 0.325 | 7.62 | 8.26 |
| E1 | 0.240 | 0.280 | 6.10 | 7.11 |
| e | 0.100 BSC | | 2.54 BSC | |
| eB | ---- | 0.430 | --- | 10.92 |
| L | 0.115 | 0.150 | 2.92 | 3.81 |
| M | ---- | 10° | --- | 10° |

STYLE 1:
PIN 1. CATHODE
2. CATHODE
3. CATHODE
4. CATHODE
5. CATHODE
6. CATHODE
7. CATHODE
8. CATHODE
9. ANODE
10. ANODE
11. ANODE
12. ANODE
13. ANODE
14. ANODE
15. ANODE
16. ANODE

STYLE 2:
PIN 1. COMMON DRAIN
2. COMMON DRAIN
3. COMMON DRAIN
4. COMMON DRAIN
5. COMMON DRAIN
6. COMMON DRAIN
7. COMMON DRAIN
8. COMMON DRAIN
9. GATE
10. SOURCE
11. GATE
12. SOURCE
13. GATE
14. SOURCE
15. GATE
16. SOURCE

## GENERIC MARKING DIAGRAM*



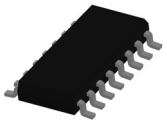

XXXXX = Specific Device Code
A = Assembly Location
WL = Wafer Lot
YY = Year
WW = Work Week
G = Pb−Free Package

*This information is generic. Please refer to device data sheet for actual part marking. Pb−Free indicator, “G” or microdot “▪”, may or may not be present. Some products may not follow the Generic Marking.

| | | |
|---|---|---|
| **DOCUMENT NUMBER:** | **98ASB42431B** | Electronic versions are uncontrolled except when accessed directly from the Document Repository. Printed versions are uncontrolled except when stamped “CONTROLLED COPY” in red. |
| **DESCRIPTION:** | **PDIP−16** | **PAGE 1 OF 1** |

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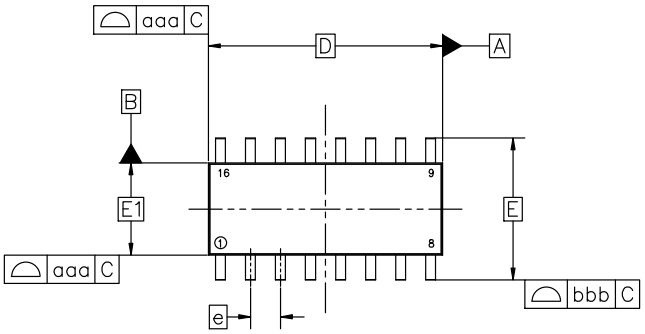
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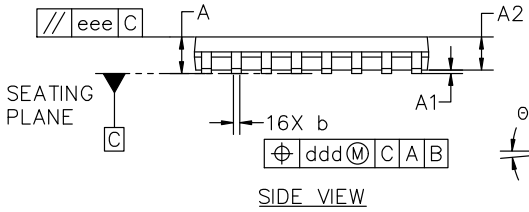
**SOIC−16 9.90x3.90x1.37 1.27P**
CASE 751B
ISSUE M

DATE 18 OCT 2024

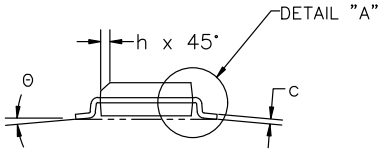
NOTES:
1. DIMENSIONING AND TOLERANCING PER ASME Y14.5M, 2018.
2. DIMENSION IN MILLIMETERS. ANGLE IN DEGREES.
3. DIMENSIONS D AND E1 DO NOT INCLUDE MOLD PROTRUSION.
4. MAXIMUM MOLD PROTRUSION 0.15mm PER SIDE.
5. DIMENSION b DOES NOT INCLUDE DAMBAR PROTRUSION. ALLOWABLE DAMBAR PROTRUSION SHALL BE 0.127mm TOTAL IN EXCESS OF THE b DIMENSION AT MAXIMUM MATERIAL CONDITION.



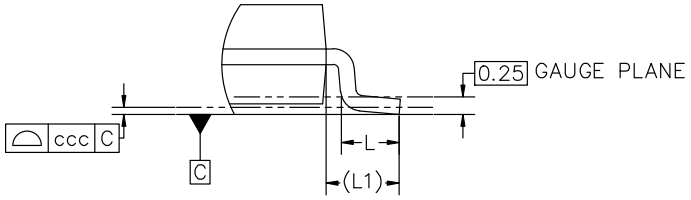

TOP VIEW




SIDE VIEW

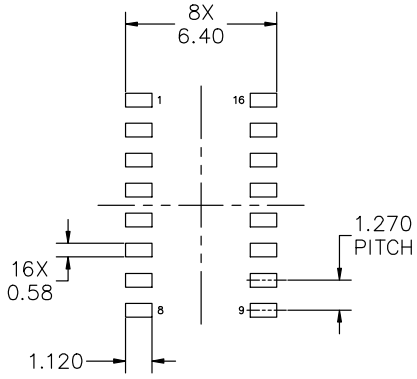



END VIEW




DETAIL "A"
Scale 3:1

| MILLIMETERS | | | |
|---|---|---|---|
| DIM | MIN | NOM | MAX |
| A | 1.35 | 1.55 | 1.75 |
| A1 | 0.10 | 0.18 | 0.25 |
| A2 | 1.25 | 1.37 | 1.50 |
| b | 0.35 | 0.42 | 0.49 |
| c | 0.19 | 0.22 | 0.25 |
| D | 9.90 BSC | | |
| E | 6.00 BSC | | |
| E1 | 3.90 BSC | | |
| e | 1.27 BSC | | |
| h | 0.25 | −−− | 0.50 |
| L | 0.40 | 0.83 | 1.25 |
| L1 | 1.05 REF | | |
| Θ | 0° | −−− | 7° |
| TOLERANCE OF FORM AND POSITION | | | |
| aaa | 0.10 | | |
| bbb | 0.20 | | |
| ccc | 0.10 | | |
| ddd | 0.25 | | |
| eee | 0.10 | | |




RECOMMENDED MOUNTING FOOTPRINT

*FOR ADDITIONAL INFORMATION ON OUR PB−FREE STRATEGY AND SOLDERING DETAILS, PLEASE DOWNLOAD THE onsemi SOLDERING AND MOUNTING TECHNIQUES REFERENCE MANUAL, SOLDERRM/D

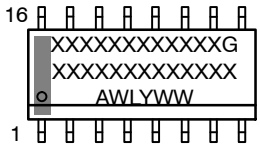
| | | | |
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| **DESCRIPTION:** | **SOIC−16 9.90X3.90X1.37 1.27P** | | **PAGE 1 OF 2** |

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**SOIC−16 9.90x3.90x1.37 1.27P**
CASE 751B
ISSUE M

DATE 18 OCT 2024

## GENERIC MARKING DIAGRAM*




XXXXX = Specific Device Code
A = Assembly Location
WL = Wafer Lot
Y = Year
WW = Work Week
G = Pb−Free Package

*This information is generic. Please refer to device data sheet for actual part marking. Pb−Free indicator, “G” or microdot “▪”, may or may not be present. Some products may not follow the Generic Marking.

STYLE 1:

| PIN | |
|---|---|
| 1. | COLLECTOR |
| 2. | BASE |
| 3. | EMITTER |
| 4. | NO CONNECTION |
| 5. | EMITTER |
| 6. | BASE |
| 7. | COLLECTOR |
| 8. | COLLECTOR |
| 9. | BASE |
| 10. | EMITTER |
| 11. | NO CONNECTION |
| 12. | EMITTER |
| 13. | BASE |
| 14. | COLLECTOR |
| 15. | EMITTER |
| 16. | COLLECTOR |

STYLE 2:

| PIN | |
|---|---|
| 1. | CATHODE |
| 2. | ANODE |
| 3. | NO CONNECTION |
| 4. | CATHODE |
| 5. | CATHODE |
| 6. | NO CONNECTION |
| 7. | ANODE |
| 8. | CATHODE |
| 9. | CATHODE |
| 10. | ANODE |
| 11. | NO CONNECTION |
| 12. | CATHODE |
| 13. | CATHODE |
| 14. | NO CONNECTION |
| 15. | ANODE |
| 16. | CATHODE |

STYLE 3:

| PIN | |
|---|---|
| 1. | COLLECTOR, DYE #1 |
| 2. | BASE, #1 |
| 3. | EMITTER, #1 |
| 4. | COLLECTOR, #1 |
| 5. | COLLECTOR, #2 |
| 6. | BASE, #2 |
| 7. | EMITTER, #2 |
| 8. | COLLECTOR, #2 |
| 9. | COLLECTOR, #3 |
| 10. | BASE, #3 |
| 11. | EMITTER, #3 |
| 12. | COLLECTOR, #3 |
| 13. | COLLECTOR, #4 |
| 14. | BASE, #4 |
| 15. | EMITTER, #4 |
| 16. | COLLECTOR, #4 |

STYLE 4:

| PIN | |
|---|---|
| 1. | COLLECTOR, DYE #1 |
| 2. | COLLECTOR, #1 |
| 3. | COLLECTOR, #2 |
| 4. | COLLECTOR, #2 |
| 5. | COLLECTOR, #3 |
| 6. | COLLECTOR, #3 |
| 7. | COLLECTOR, #4 |
| 8. | COLLECTOR, #4 |
| 9. | BASE, #4 |
| 10. | EMITTER, #4 |
| 11. | BASE, #3 |
| 12. | EMITTER, #3 |
| 13. | BASE, #2 |
| 14. | EMITTER, #2 |
| 15. | BASE, #1 |
| 16. | EMITTER, #1 |

STYLE 5:

| PIN | |
|---|---|
| 1. | DRAIN, DYE #1 |
| 2. | DRAIN, #1 |
| 3. | DRAIN, #2 |
| 4. | DRAIN, #2 |
| 5. | DRAIN, #3 |
| 6. | DRAIN, #3 |
| 7. | DRAIN, #4 |
| 8. | DRAIN, #4 |
| 9. | GATE, #4 |
| 10. | SOURCE, #4 |
| 11. | GATE, #3 |
| 12. | SOURCE, #3 |
| 13. | GATE, #2 |
| 14. | SOURCE, #2 |
| 15. | GATE, #1 |
| 16. | SOURCE, #1 |

STYLE 6:

| PIN | |
|---|---|
| 1. | CATHODE |
| 2. | CATHODE |
| 3. | CATHODE |
| 4. | CATHODE |
| 5. | CATHODE |
| 6. | CATHODE |
| 7. | CATHODE |
| 8. | CATHODE |
| 9. | ANODE |
| 10. | ANODE |
| 11. | ANODE |
| 12. | ANODE |
| 13. | ANODE |
| 14. | ANODE |
| 15. | ANODE |
| 16. | ANODE |

STYLE 7:

| PIN | |
|---|---|
| 1. | SOURCE N-CH |
| 2. | COMMON DRAIN (OUTPUT) |
| 3. | COMMON DRAIN (OUTPUT) |
| 4. | GATE P-CH |
| 5. | COMMON DRAIN (OUTPUT) |
| 6. | COMMON DRAIN (OUTPUT) |
| 7. | COMMON DRAIN (OUTPUT) |
| 8. | SOURCE P-CH |
| 9. | SOURCE P-CH |
| 10. | COMMON DRAIN (OUTPUT) |
| 11. | COMMON DRAIN (OUTPUT) |
| 12. | COMMON DRAIN (OUTPUT) |
| 13. | GATE N-CH |
| 14. | COMMON DRAIN (OUTPUT) |
| 15. | COMMON DRAIN (OUTPUT) |
| 16. | SOURCE N-CH |




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